Exploratory Data Analysis: NOMA SIC Dataset

Source file: `1792033912881_noma_sic_dataset__1_.csv` **Size:** 60,000 rows × 8 columns (2.7 MB)

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1. Overview

The dataset looks like the output of a simulation of a two-user non-orthogonal multiple access (NOMA) link with successive interference cancellation (SIC). Each row is one simulated trial, with channel SNRs for a near and a far user, a pilot SNR, the power-allocation coefficient for the near user, and an outcome indicating whether SIC failed and what goodput resulted.

Column	Type	Range / Levels	Interpretation (inferred)
`near_snr_db`	float	0.0 – 35.0	SNR of the near user (dB)
`far_snr_db`	float	0.0 – 25.0	SNR of the far user (dB)
`gap_db`	float	0.0 – 34.9	Channel gap between users, equal to |near − far|
`pilot_snr_db`	float	10.0 – 25.0	Pilot SNR used for channel estimation (dB)
`a_near`	float	{0.15, 0.25, 0.35}	Power allocation fraction for the near user
`R_far`	float	constant 1.5	Target rate for the far user
`sic_failure`	int	{0, 1}	Binary target: SIC failed (1) or not (0)
`goodput`	float	{0.0, 1.5}	Achieved goodput; 1.5 if no failure, 0 if failure

Data quality: no missing values, no duplicate rows, all numeric types as expected.

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2. Key Findings

1. **SIC failure is rare.** Only 1,898 of 60,000 trials (3.16%) failed. The classes are heavily imbalanced, so accuracy is a misleading metric here. A model predicting "no failure" everywhere scores 96.8%.
2. **Far-user SNR is the strongest driver of failure.** Failure rate falls from about 10.3% when far SNR is below 3 dB to 0.7% above 12 dB, and to 0.2% in the top bin (21.9–25 dB).
3. **Pilot SNR and near SNR also matter, in the expected direction.** Failure rate drops from about 6.9% at the lowest pilot SNR bin to under 1% at the highest, and from about 5.4–6.0% at low near SNR to about 0.8% at high near SNR.

4. **The power-allocation coefficient has a weak effect.** Failure rises slightly with `a_near`: 2.5% at 0.15, 3.3% at 0.25, 3.7% at 0.35. This is the expected direction, since more power to the near user leaves less for the far user to decode, but the effect is small relative to the SNR effects.
5. **Failure is predictable from the SNRs but not deterministic.** A logistic regression on the four continuous and power-allocation features reaches a cross-validated ROC AUC of 0.856. The standardized coefficients rank the features by influence: far SNR (−1.34), near SNR (−0.68), pilot SNR (−0.78), and `a_near` (+0.18).
6. **`goodput` and `sic_failure` are redundant.** Their correlation is exactly −1, and goodput equals 1.5 × (1 − sic_failure). Using goodput as a feature to predict failure leaks the target.

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3. Univariate Analysis

Numeric features

Feature	Mean	Std	Min	Median	Max
near_snr_db	17.49	10.09	0.00	17.51	35.00
far_snr_db	12.50	7.22	0.00	12.54	25.00
gap_db	10.92	7.66	0.00	9.65	34.92
pilot_snr_db	17.52	4.33	10.00	17.51	25.00

The SNR features have means near the midpoints of their ranges and standard deviations close to the expected value for uniform distributions (uniform on [0, 35] gives SD ≈ 10.1; uniform on [0, 25] gives ≈ 7.2; uniform on [10, 25] gives ≈ 4.3). This suggests the inputs were sampled uniformly over these ranges, which is typical of synthetic simulation data.

Categorical and constant features

- `a_near` takes three levels in near-equal proportions: 0.15 (20,087 rows), 0.25 (19,829), 0.35 (20,084).
- `R_far` is constant at 1.5 in all rows, so it carries no information and can be dropped for modeling.
- `sic_failure`: 58,102 zeros and 1,898 ones (3.16% positive).
- `goodput` takes only two values, 1.5 and 0.0, so it is effectively a binary variable.

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4. Bivariate Analysis

Correlation with `sic_failure`

Feature	Correlation
far_snr_db	−0.176
pilot_snr_db	−0.116
near_snr_db	−0.103
gap_db	−0.071
a_near	+0.027

Failure rate by binned feature

far_snr_db bin	Failure rate
0 – 3.1	10.3%
3.1 – 6.3	6.5%
6.3 – 9.4	3.9%
9.4 – 12.5	2.2%
12.5 – 15.6	1.2%
15.6 – 18.8	0.7%
18.8 – 21.9	0.4%
21.9 – 25.0	0.2%

The relationship is monotonic and steep at low far SNR, then flattens. Failures are concentrated in the weak-far-user region.

Gap and near SNR. `gap_db` is correlated with `near_snr_db` (0.45) and negatively with `far_snr_db` (−0.26). These correlations come from the definition of the gap, not from independent information, so `gap_db` should not be treated as an independent predictor alongside the two SNRs.

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5. Data Consistency Checks

- **`gap_db` is derived.** It equals |near_snr_db − far_snr_db| to within 0.0001 (rounding). This confirms it is a computed column.
- **`goodput` is derived from `sic_failure`.** Goodput is exactly 1.5 when `sic_failure` = 0 and exactly 0 when `sic_failure` = 1.

- **No constant leakage beyond the above.** `R_far` is constant, as noted.

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6. Modeling Implications

- **Target:** `sic_failure` (binary, 3% positive). Use stratified splits, class weights or resampling, and report precision, recall, F1, and ROC AUC or PR AUC rather than accuracy.
- **Features to use:** `near_snr_db`, `far_snr_db`, `pilot_snr_db`, and `a_near`. Consider whether to include `gap_db`, since it is a deterministic function of the two SNRs and adds no new information for most models.
- **Features to exclude:** `goodput` (direct leakage of the target) and `R_far` (constant).
- **Expected difficulty:** A linear logistic model already gets AUC ≈ 0.86. Nonlinear models may improve on this, but the failure rate is driven by a probabilistic process, so there is likely a ceiling on achievable performance.

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7. Caveats and Open Questions

- The dataset's origin is not documented. The uniform sampling and the clean derived columns suggest a simulator. If so, the findings describe the simulator's assumptions and may not transfer to real channels.
- The meaning of the pilot SNR's role is not stated. It is assumed here to affect channel estimation quality and therefore SIC reliability.
- The threshold rule that produces `sic_failure` is not in the data. The failure probability looks smooth in each feature, which suggests a stochastic decoding model rather than a hard threshold on one variable.
- Without a data dictionary, the units and definitions of `a_near` and `R_far` are inferred from NOMA conventions.

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8. Recommended Next Steps

1. Confirm the data source and column definitions, especially for `a_near`, `R_far`, and the failure criterion.
2. Check for interactions, particularly between `far_snr_db` and `a_near`, since power allocation should matter most when the far user is weak.
3. Fit a gradient-boosted model and compare its AUC against the logistic baseline of 0.856.
4. Drop `goodput` and `gap_db` from the feature set before modeling, and document why.